

# Perdiction Results

2024-10-02

## Data Summary:

- **Number of observations for bill passage predictions:** 40,000.
- **Number of observations for bill summary completion:** 39,997, 3 null responses were also dropped. A total of 38 responses exceeded the 750-token limit, and 31 responses modified the beginning of the provided bill summary, but were still included in the analysis.

## Results

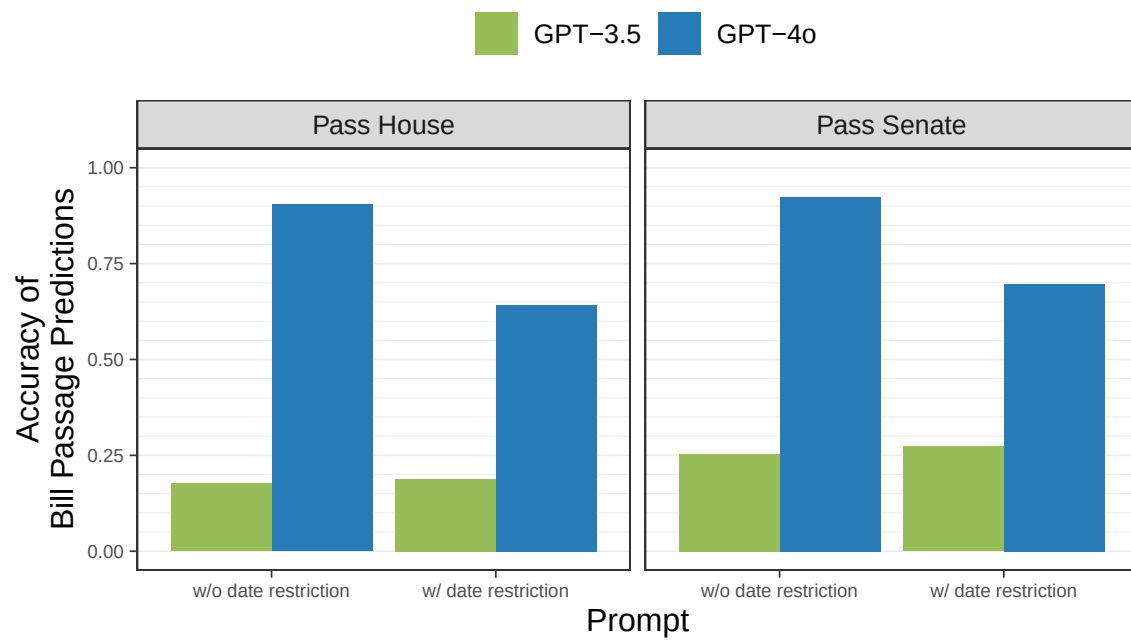
### Bill Passage

Table 1: True Bill Passage Rates

| Chamber     | True Passage Rate |
|-------------|-------------------|
| Pass House  | 0.081             |
| Pass Senate | 0.061             |

Table 2: Performance Evaluation of LLM Predictions of Bill Passage

| Predicted Outcome | Model   | Prompt               | Accuracy | TPR   | FPR   |
|-------------------|---------|----------------------|----------|-------|-------|
| Pass House        | GPT-3.5 | w/o date restriction | 0.176    | 0.981 | 0.895 |
| Pass House        | GPT-3.5 | w/ date restriction  | 0.189    | 0.980 | 0.881 |
| Pass House        | GPT-4o  | w/o date restriction | 0.905    | 0.174 | 0.031 |
| Pass House        | GPT-4o  | w/ date restriction  | 0.641    | 0.678 | 0.362 |
| Pass Senate       | GPT-3.5 | w/o date restriction | 0.253    | 0.982 | 0.794 |
| Pass Senate       | GPT-3.5 | w/ date restriction  | 0.273    | 0.980 | 0.773 |
| Pass Senate       | GPT-4o  | w/o date restriction | 0.924    | 0.222 | 0.030 |
| Pass Senate       | GPT-4o  | w/ date restriction  | 0.696    | 0.725 | 0.306 |



## Bill Summary Completion

Table 3: Performance Evaluation of LLM Completion of Bill Summaries (Non-Embedded)

| Model   | Prompt               | Average |
|---------|----------------------|---------|
| GPT-3.5 | w/o date restriction | 0.017   |
| GPT-3.5 | w/ date restriction  | 0.019   |
| GPT-4o  | w/o date restriction | 0.044   |
| GPT-4o  | w/ date restriction  | 0.043   |

Table 4: Performance Evaluation of LLM Completion of Bill Summaries (Embedded)

| Metric             | Model   | Prompt               | Average | Baseline | Relative to Baseline |
|--------------------|---------|----------------------|---------|----------|----------------------|
| Cosine Similarity  | GPT-3.5 | w/o date restriction | 0.488   | 0.229    | 1.131                |
| Cosine Similarity  | GPT-3.5 | w/ date restriction  | 0.498   | 0.229    | 1.177                |
| Cosine Similarity  | GPT-4o  | w/o date restriction | 0.535   | 0.229    | 1.334                |
| Cosine Similarity  | GPT-4o  | w/ date restriction  | 0.540   | 0.230    | 1.352                |
| Euclidean Distance | GPT-3.5 | w/o date restriction | 0.986   | 1.240    | 0.205                |
| Euclidean Distance | GPT-3.5 | w/ date restriction  | 0.974   | 1.240    | 0.214                |
| Euclidean Distance | GPT-4o  | w/o date restriction | 0.921   | 1.240    | 0.257                |
| Euclidean Distance | GPT-4o  | w/ date restriction  | 0.916   | 1.240    | 0.261                |

